

Double hashing

iteratively check the hash value of k .

$$\text{hash}_1(k) + j \cdot \text{hash}_2(k)$$

$$\text{hash}_1(k) = k \bmod 13$$

$$\text{hash}_2(k) = 7 - k \bmod 7$$

$$\text{Keys} = (18, 41, 22, 44)$$

k	hash_1	hash_2
18	5	3
41	2	1
22	9	6
44	5	5

1		41			18			25	44				
0	1	2	3	4	5	6	7	8	9	10	11	12	13

$$5 + 0 \times 3 = 5$$

$$2 + 0 \times 1 = 2$$

$$9 + 0 \times 6 = 9$$

$$5 \times 0 + 5 = 10$$

$j=0 \rightarrow$ until index is empty
and I can fill that index

Rehashing

Load Factor

$\rightarrow n$ entries

$\rightarrow n$ buckets

Load Factor or

$$\lambda = \frac{n}{N}$$

Preferable $\lambda < 1$ i.e. $N > n$

When $A > 1$, increase the
number of buckets.

⇒ Rehashing

N' ↳ Increase no. of buckets

↳ modify hash function

$x \bmod N$ → change to

$x \bmod N'$

Apply $h'(x)$ (modified
hash function) to existing
elements

N' ? closest prime number
to $2N$

$$h(x) = x \bmod 3$$

keys = (6, 7, 8)

6	7	8
0	1	2

$$n = 3$$

$$\lambda = 1$$

$$N = 3$$

$$N' = 7$$

$$h'(x) = x \bmod 7$$

7	8					6
0	1	2	3	4	5	6

$$n = 3$$

$$\lambda = \frac{3}{7} < 1$$

$$N = 7$$